

Attachment Rescue: giving NCI's buried supplementary datasets a permanent home

Submitted by Ochiai, Ojio · Individual · Narrative, four prompts plus disclosure

PROMPT 1

Significance and Approach

The gap. Before the 2023 NIH Data Management and Sharing Policy, the normal way to share a cancer dataset was to attach it to the paper: gene expression tables, patient cohort tables, drug screens and mass spectrometry measurements, saved as Excel files and uploaded as "Supplementary Data 3." Those files still exist in PubMed Central, but no dataset catalogue lists them and nothing describes what is inside. A researcher looking for glioblastoma cohort data with survival and IDH status cannot find a 121-patient table that has sat in a supplementary file since 2018 without already knowing the paper.

How big it is. We measured it. Europe PMC lists 58,469 NCI-funded open-access papers from 2016 to 2022; 26,228 have supplementary files and no text-mined reference to any of ten data repositories (GEO, SRA, dbGaP, EGA, ArrayExpress, PRIDE, BioProject, MetaboLights, PDB, GenBank). We drew random samples of 300 papers for each of four cancers (glioblastoma, breast, colorectal, leukemia), downloaded every attachment, and opened every spreadsheet, CSV, Word and PDF file. A file counts as a dataset only if its largest table has at least 50 rows and 3 columns, or 1,000 rows. **The four cancers agree within three points: pooled, 202 of 1,076 papers (19%) hold a data table and 156 (14.5%, confidence interval 12% to 17%) hold it as a native spreadsheet rather than a table inside a PDF.** 167 of the 202 (83%) are CC BY or CC0, so their files can be re-deposited today. Tables ran up to 152,683 rows. The classifier's decisions were re-examined on 90 files (with the AI assistant, not blind; verdicts published): all 60 it called datasets were real data tables, and every miss was under 50 rows. A blind human check is next. We then ran a census of the full population from each paper's full-text attachment list, downloading nothing: 73,795 attachments across 25,138 readable papers. **4,759 papers (18.9%) carry at least one spreadsheet, CSV or tab-delimited text attachment, 15,458 such files in total, 15.9 gigabytes of Excel alone.** 3,824 of those papers (80%) are licensed CC BY or CC0. After discounting papers that cite a repository our filter missed (15%, checked on 500 papers) and files already on Figshare (13.5%, verified file by file), the estimate is roughly 2,800 to 3,900 papers holding 8,000 to 11,000 buried dataset files, about three-quarters under a license that permits re-deposit. The

authors say so themselves: in the same 500 papers, 17% of data-availability statements say the data is "in the paper or supplementary files" and 15% say "on request." And it is not history: in a random 600 NCI open-access papers from 2024 and 2025, under the new policy, 23.2% still carry a spreadsheet attachment with no repository accession, higher than before.

Who is affected. Researchers, because data they need is unfindable; trainees and small labs most of all, because they lack the network that knows which paper to open. Students, patient advocates and Data Jamboree participants, who meet a literature that hides its data in attachments. The patients: verified patient-level tables in our four samples alone describe about 4,400 patient records, roughly 500 from institutional cohorts we could not find in TCGA, GEO, dbGaP or any catalogue. The authors, whose data earns no citations because it has no identifier. And NCI, which paid for these datasets twice and receives no reuse in return.

Why it matters to NCI and ODS goals. These papers were compliant. They followed the norms of their time, and the data still vanished from view: the situation the challenge describes when it says policy alone does not drive culture change. The challenge FAQ says a Track 1 "proposal to develop a mechanism for making outputs publicly available is explicitly responsive"; this is that mechanism, for outputs NCI already paid for. Data only insiders can find is not broadly, rapidly or equitably shared. ODS's own workshop report names "incomplete, incorrect, or missing metadata" and the findability of siloed data as major bottlenecks; its Data Jamboree paper names "limited awareness of available datasets" as a primary barrier. A supplementary file is the purest case of both: real data with no metadata, in a silo of one.

The approach. A retrospective rescue over infrastructure that already exists:

1. **Find.** Query Europe PMC for NCI-funded open-access papers with supplementary files and no repository accession; read each paper's full text to list its attachments. One call per paper, no downloads.
2. **Sort.** Download the tabular attachments and classify each: dataset, small table, or not data. Separate primary measurements (cohort tables, expression tables, counts, spectra) from derived results (gene lists, pathway tables). Skip peer-review files, reporting summaries and figures.
3. **Describe.** Generate a standard dataset record for each, in the DataCite format repositories and search engines read: title, creators and license from the article, article DOI and PMC identifier, funder, format, sheet names, and a one-sentence plain-language description of what the table contains. A person spot-checks.
4. **Deposit or register.** For CC BY and CC0 articles, deposit the unmodified file in one of the free general-purpose repositories NIH already recognizes, which assigns it a permanent identifier (a DOI). For no-derivatives licenses, and for any file whose

authors object, register a metadata record pointing to the file's existing location in PubMed Central instead of copying it. Who deposits, and how authors are told, is addressed below.

5. **List.** Deliver the rescued records to the Index of NCI Studies as a curated source file in the format its pipeline already accepts. We have produced that file for the 462 rescued datasets, in the index's exact 27-column schema; adding them is one load, not new infrastructure.

6. **Keep alive.** Check every rescued link monthly and publish the result.

Steps 1 through 3 have run on 1,200 papers across four cancers, and step 4 as far as packaging: eight glioblastoma datasets sit ready for deposit with complete metadata, awaiting the governance decision below. We also put three rescued glioma tables back to work: two TCGA-derived tables from different papers join cleanly on 244 shared patients and, pooled to 1,037, reproduce the known IDH survival split; the third is a 121-patient hospital cohort found in no catalogue. The proposal is to run the whole pipeline over the full NCI corpus, cancer by cancer.

PROMPT 2

Potential Impact on the Cancer Research Community

Immediate. Thousands of existing cancer datasets become findable, citable and described: by the rates above, 8,000 to 11,000 in the NCI corpus alone, about three-quarters freely re-depositable. No new data is needed; the cost of finding them is all that stands between them and reuse.

Breadth. The gap is not specific to any cancer or to cancer itself. The same filename census on 300-paper samples from five other institutes gives NHLBI 15%, NIDDK 18%, NIMH 19%, NIA 21%, NINDS 24%, against 18% for NCI; those five add 63,000 papers to the population. Because the pipeline is defined by a funder tag and a date range, any NCI Division, Office or Center, or any NIH institute, can run it unchanged.

For non-data scientists. A rescued dataset arrives with a plain-language description and a stable identifier: a community college instructor can hand students a real patient cohort table with a DOI, and a Data Jamboree team can search for "cohort with survival data" and get a hit. We propose offering the rescued collection to the November 2026 NCI Data Jamboree, an event ODS runs for exactly the users the challenge names.

For authors and for credit. Each rescued dataset gains an identifier linked to its article, so invisible data can start to accrue citations; papers whose data statements link to a repository receive up to 25% more citations (Colavizza et al., 2020), a correlation, but one authors get here at no effort.

For policy and infrastructure across NCI. The census gives ODS something it does not have: a per-year, per-journal account of how much NCI-funded data was shared only as attachments and how much is legally recoverable (Nature Communications alone: 889 papers, 3,201 files). By funding mechanism, in 8,040 census papers checked, 47% acknowledge an R01-type grant, 45% a P30 Cancer Center grant, 18% a cooperative agreement or consortium, 17% a training award and 9% a SPORE, so the buried data spans the extramural divisions, the Cancer Centers Program and the training portfolio, and the census can be cut by any of them. That informs guidance, lets program officers report the public benefit of past investments concretely, and gives the Index of NCI Studies its first source of datasets drawn from publications. From 1 October 2026 grantees must report data-sharing progress in each progress report (NOT-OD-26-100); a census re-runnable per grant gives program staff an independent check.

PROMPT 3

Innovation and Awareness of Existing Efforts

What exists. - **Large-scale processing of supplementary files has been done.** NLM's FAIR-SMART (Wei, Lu and colleagues, PLOS Biology, 2025) aggregates PubMed Central supplementary files, converts them to a machine-readable form, uses language models to categorize their tables, and serves them through an API. Gobeill and colleagues (Bioinformatics Advances, 2025) processed 36.2 million supplementary files and built a text search index; the Europe PMC team (Kafkas et al., 2015) mined supplements for accession numbers earlier. These are retrieval systems for text mining and search. None decides which files are datasets in the sense a researcher means, describes them in plain language, gives them a dataset identifier, scopes them to a funder, or lists them in a discovery catalogue. FAIR-SMART is a natural upstream for rescue, not a substitute.

- **The files are already mirrored, as files.** EMBL-EBI's BioStudies holds one record per Europe PMC article (for example S-EPMC6193287) listing its supplementary files under the article title; PLOS assigns component DOIs to supporting files, and Springer Nature and BMC mirror theirs to Figshare. All are per-article file dumps: no record says which file is a dataset, what it contains or who is in it, none carries a dataset identifier, and none reaches NCI's index. Mirroring is the storage half; rescue is the description and catalogue half. We checked our census against Figshare: files from BMC and other Springer Nature journals, 13.5% of census spreadsheet papers when verified file by file, already have a Figshare identifier; files from Nature Communications, Scientific Reports, eLife, Oncotarget and Cell Press journals, the large majority, do not. Even the mirrored files have no dataset description and do not reach NCI's index, so they need the sorting, describing and listing steps but not the deposit step.

- **The scale of the problem is documented.** Federer et al. (2018): 69% of PLOS ONE data availability statements pointed to the paper or supplements, 18% to a repository; Colavizza et al.

(2020): repository links in 21% of PLOS and 12% of BMC papers; Federer (2022): DOIs outlast plain links, which is why rescued files get one. - **The catalogue side exists.** The Index of NCI Studies, managed by ODS and built by CBIIT, ingests datasets from GEO, SRA, dbGaP and several curated NCI sources through a public pipeline, but nothing from general-purpose repositories or publications. Its team already runs automated enrichment, including a published tool that maps study text to disease codes, so a curated source of publication-derived datasets is a next step, not a new capability. NIH's Generalist Repository Ecosystem Initiative has no retrospective harvest component.

What is new. The combination, and the direction. We found no effort that has taken a funder's back catalogue, classified which attachments are datasets, described them, given them identifiers, and delivered them into the funder's own catalogue. Extraction and mirroring exist; classification, deposit and listing do not. Every component is standard; what is new is pointing them backwards at data shared in good faith and lost anyway.

What we are not proposing. Not a new repository, catalogue or metadata standard, and not a replacement for the 2023 policy; rescue is the backfill for everything published before it, and the monitor for what still slips through.

PROMPT 4

Transferability, Sustainability, and Feasibility

Path to adoption. Every component is public and run by someone else: Europe PMC and FAIR-SMART serve the papers and attachments, general-purpose repositories assign identifiers for free, and the Index of NCI Studies pipeline is open source and ingests curated source files. Adoption is one new curated source in that pipeline, a decision inside NCI rather than new infrastructure.

What it would take. Seconds of compute and cents of storage per paper; the census needs no downloads, and rescuing a cancer the size of glioblastoma is a few hundred files. Human time goes to spot-checking metadata and one conversation with the INS team about the source file, which already exists.

Constraints, stated plainly.

- *Licensing.* 80% of the spreadsheet-bearing papers are CC BY or CC0 and their files can be deposited unchanged. About 9% are CC BY-NC-ND, which forbids modified copies; those get a metadata record pointing at PubMed Central. Author manuscripts outside the open-access subset are excluded. License tags are enforced before anything is copied.

- *Who deposits, and author consent.* No third party silently uploads authors' files. Preferred: NCI or ODS acts as depositor under an institutional repository account, as NIH does for NIH-authored manuscripts in PubMed Central; otherwise authors are notified 60 days before deposit with a one-click opt-out, and objections yield a metadata-only record. Either way the record names the original authors as creators, links the article, carries a route for correction or removal, and NCI never owns the data. Tables describing human participants, even de-identified, get a review step before copying.
- *Signal versus noise.* About two-thirds of sampled papers had no tabular data at all; the classifier keeps that noise out, and its accuracy is published. About half of rescued tables are derived results rather than raw measurements, and each record says which.
- *Duplication.* Some attachments duplicate deposited data or already carry a Figshare identifier; records are matched by article DOI and filename before deposit and an existing identifier is reused.
- *The INS intake.* The one dependency on others. If INS does not add the source, the datasets still exist with DOIs in repositories NIH recognizes, and the census still stands.

Why it persists. After the backlog, the pipeline runs quarterly on new papers and monthly on link health, both scheduled checks. What outlives any team is the curated source file and the census, both public and reproducible.

DISCLOSURE

Use of Generative AI

Generative AI (Anthropic's Claude) was used to help write and run the analysis code, to help draft and edit this narrative, and to draft the dataset descriptions. All sampling, classification and counts come from scripts included with the supporting materials and can be rerun from public data; the re-examination of 90 classifier decisions was done with the assistant and is published for audit. All decisions about scope, claims and recommendations are the submitter's own, who takes full responsibility for the accuracy of the content.